

WELL COMPLETION REPORT (SYNTHETIC)

OIL-RDS-013 - Rudrasagar-013

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000150
Field / District	Rudrasagar / Sivasagar
Block	PEL-ASSAM-06
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2009-08-01 / 2009-09-21
Measured / TVD	2642 m / 2614 m
Trajectory	Vertical; inc 4.5 deg; az 85.7 deg
Formation Interval	1583-2406 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3265 psi
Fracture Gradient	15.34 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (50.5/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	660	Surface
9 5/8 in casing	1532	Intermediate
7 in liner/casing	2324	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2009-08-26 1403 m	Pressure Spike	Critical	Circulating pressure climbed without equivalent torque change. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Correlate pressure anomalies with flow before escalating risk.
2009-08-21 1170 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2009-09-14 2303 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2009-09-17 2443 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.
2009-08-19 1084 m	Torque Spike	Medium	Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2009-08-22 1191 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2009-08-24 1297 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 2642 m MD in the Barail interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.